



Millennium AL (AFCO 4388)

Safety Data Sheet

according to Federal Register / Vol. 77, No. 58 / Monday, March 26, 2012 / Rules and Regulations

Revision Date: 12/06/2024

Version: 1.0

SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY

Product Identifier

Product Form: Mixture

Product Name: Millennium AL

Product Code: AFCO 4388

Intended Use of the Product

Use of the Substance/Mixture: Detergent for use on food processing equipment. For professional use only.

Name, Address, and Telephone of the Responsible Party

Company

AFCO

550 Development Avenue

Chambersburg, PA 17201

T: 800-345-1329

www.afcocare.com

Emergency Telephone Number

Emergency Number : 1-800-424-9300 (CHEMTREC)

SECTION 2: HAZARDS IDENTIFICATION

Classification of the Substance or Mixture

Classification (GHS-US)

PRODUCT as SOLD

Ox. Liq. 1	H271
Org. Perox. D	H242
Met. Corr. 1	H290
Acute Tox. 4 (Oral)	H302
Acute Tox. 4 (Inhalation:dust,mist)	H332
Skin Corr. 1A	H314
Eye Dam. 1	H318
STOT SE 3	H335
Aquatic Acute 2	H401

PROUDUCT at USE DILUTION

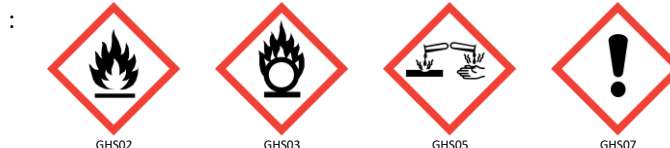
Skin Irrit. 3	H316
Eye Irrit. 2B	H320

PRODUCT as SOLD

Label Elements

GHS-US Labeling

Hazard Pictograms (GHS-US)



Signal Word (GHS-US)

: Danger

Hazard Statements (GHS-US)

: H242 - Heating may cause a fire
H271 - May cause fire or explosion; strong oxidizer
H290 - May be corrosive to metals
H302+H332 - Harmful if swallowed or if inhaled
H314 - Causes severe skin burns and eye damage
H318 - Causes serious eye damage
H335 - May cause respiratory irritation
H401 - Toxic to aquatic life

Precautionary Statements (GHS-US) : P201 - Obtain special instructions before use.

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P202 - Do not handle until all safety precautions have been read and understood.
P210 - Keep away from heat, sparks, open flames, hot surfaces. - No smoking.
P220 - Keep/Store away from clothing, combustible materials, incompatible materials.
P221 - Take any precaution to avoid mixing with combustible materials, incompatible materials.
P234 - Keep only in original container.
P260 - Do not breathe vapors, mist, spray.
P264 - Wash hands, forearms, and other exposed areas thoroughly after handling.
P270 - Do not eat, drink or smoke when using this product.
P271 - Use only outdoors or in a well-ventilated area.
P273 - Avoid release to the environment.
P280 - Wear protective gloves, protective clothing, eye protection, face protection.
P283 - Wear fire/flame resistant/retardant clothing.
P301+P312 - IF SWALLOWED: Call a POISON CENTER or doctor/physician if you feel unwell.
P301+P330+P331 - IF SWALLOWED: rinse mouth. Do NOT induce vomiting.
P303+P361+P353 - IF ON SKIN (or hair): Remove/Take off immediately all contaminated clothing. Rinse skin with water/shower.
P304+P340 - IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing.
P305+P351+P338 - If in eyes: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
P306+P360 - If on clothing: Rinse immediately contaminated clothing and skin with plenty of water before removing clothes.
P308+P313 - If exposed or concerned: Get medical advice/attention.
P310 - Immediately call a POISON CENTER or doctor/physician.
P312 - Call a POISON CENTER/doctor/physician if you feel unwell.
P321 - Specific treatment (see section 4).
P330 - If swallowed, rinse mouth.
P363 - Wash contaminated clothing before reuse.
P370+P378 - In case of fire: Use appropriate media for extinction.
P371+P380+P375 - In case of major fire and large quantities: Evacuate area. Fight fire remotely due to the risk of explosion.
P390 - Absorb spillage to prevent material damage.
P403+P233 - Store in a well-ventilated place. Keep container tightly closed.
P405 - Store locked up.
P406 - Store in corrosive resistant container with a resistant inner liner.
P410 - Protect from sunlight.
P411+P235 - Store at temperatures not exceeding storage and handling temperatures. Keep cool.
P420 - Store away from other materials.
P501 - Dispose of contents/container to local, regional, national, territorial, provincial, and international regulations.

PRODUCT at USE DILUTION

Hazard Pictograms (GHS-US) : None

Signal Word (GHS-US) : Warning.

Hazard Statements (GHS-US) : H316 - Causes mild skin irritation
H320 - Causes eye irritation

Precautionary Statements (GHS-US) : P264 - Wash hands, forearms, and other exposed areas thoroughly after handling.
P305+P351+P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
P332+P313 - If skin irritation occurs: Get medical attention.
P337+P313 - If eye irritation persists: Get medical advice/attention.

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Other Hazards

Other Hazards Not Contributing to the Classification: Exposure may aggravate those with pre-existing eye, skin, or respiratory conditions. When heated to decomposition, emits toxic fumes. Contains an oxidizing material which may accelerate fire.

Unknown Acute Toxicity (GHS-US): Not available.

SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

Substances

Mixture

PRODUCT as SOLD

Name	Product identifier	% (w/w)	Classification (GHS-US)
Water	(CAS No) 7732-18-5	50 - 70	Not classified
Hydrogen peroxide	(CAS No) 7722-84-1	20 - 30	Ox. Liq. 1, H271 Acute Tox. 4 (Oral), H302 Acute Tox. 4 (Dermal), H312 Acute Tox. 4 (Inhalation: vapour), H332 Skin Corr. 1A, H314 STOT SE 3, H335 Aquatic Acute 3, H402 Aquatic Chronic 3, H412
Acetic acid	(CAS No) 64-19-7	5 - 10	Flam. Liq. 3, H226 Skin Corr. 1A, H314 Eye Dam. 1, H318 Aquatic Acute 3, H402
Peroxyacetic acid	(CAS No) 79-21-0	5 - 10	Flam. Liq. 3, H226 Org. Perox. D, H242 Acute Tox. 4 (Oral), H302 Acute Tox. 4 (Dermal), H312 Acute Tox. 4 (Inhalation), H332 Skin Corr. 1A, H314 Aquatic Acute 1, H400
Sulfuric acid	(CAS No) 7664-93-9	0.1 - 1.0	Met. Corr. 1, H290 Skin Corr. 1A, H314 Eye Dam. 1, H318 Carc. 1A, H350
Benzenesulfonic acid, C10-16-alkyl derivatives, sodium salts	(CAS No) 68081-81-2	0.1 - 1.0	Acute Tox. 4 (Oral), H302 Skin Irrit. 2, H315 Eye Dam. 1, H318 STOT SE 3, H335

PRODUCT at USE DILUTION

Name	Product identifier	% (w/w)	Classification (GHS-US)
Acetic acid	(CAS No) 64-19-7	0.095	Flam. Liq. 3, H226 Skin Corr. 1A, H314 Eye Dam. 1, H318 Aquatic Acute 3, H402
Peroxyacetic acid	(CAS No) 79-21-0	0.05	Flam. Liq. 3, H226 Org. Perox. D, H242 Acute Tox. 4 (Oral), H302 Acute Tox. 4 (Dermal), H312 Acute Tox. 4 (Inhalation), H332 Skin Corr. 1A, H314 Aquatic Acute 1, H400

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Hydrogen peroxide	(CAS No) 7722-84-1	0.21	Ox. Liq. 1, H271 Acute Tox. 4 (Oral), H302 Acute Tox. 4 (Dermal), H312 Acute Tox. 4 (Inhalation: vapour), H332 Skin Corr. 1A, H314 STOT SE 3, H335 Aquatic Acute 3, H402 Aquatic Chronic 3, H412
Sulfuric acid	(CAS No) 7664-93-9	0.007	Met. Corr. 1, H290 Skin Corr. 1A, H314 Eye Dam. 1, H318 Carc. 1A, H350

Full text of H-phrases: see section 16.

SECTION 4: FIRST AID MEASURES

PRODUCT as SOLD:

General: Never give anything by mouth to an unconscious person. If you feel unwell, seek medical advice (show the label where possible). IF exposed or concerned: Get medical advice/attention.

Inhalation: When symptoms occur: go into open air and ventilate suspected area. Remove to fresh air and keep at rest in a position comfortable for breathing. Immediately call a POISON CENTER or doctor/physician.

Skin Contact: Remove contaminated clothing. Drench affected area with water for at least 15 minutes. Immediately call a POISON CENTER or doctor/physician.

Eye Contact: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Immediately call a POISON CENTER or doctor/physician.

Ingestion: Rinse mouth. Do NOT induce vomiting. Immediately call a POISON CENTER or doctor/physician.

PRODUCT at USE DILUTION:

General: Never give anything by mouth to an unconscious person. If you feel unwell, seek medical advice (show the label where possible). IF exposed or concerned: Get medical advice/attention.

Inhalation: Remove to fresh air. Get medical attention if symptoms occur.

Skin Contact: Rinse with plenty of water.

Eye Contact: Rinse with plenty of water

Ingestion: Rinse mouth. Get medical attention if symptoms occur.

Most Important Symptoms and Effects Both Acute and Delayed

PRODUCT as SOLD:

General: Harmful if swallowed. Harmful if inhaled. Corrosive. Causes burns. Causes serious eye damage.

Inhalation: Harmful if inhaled. May cause respiratory irritation. Inhalation may cause immediate severe irritation progressing quickly to chemical burns.

Skin Contact: Contact may cause immediate severe irritation progressing quickly to chemical burns.

Eye Contact: Causes serious eye damage.

Ingestion: Swallowing a small quantity of this material will result in serious health hazard. Harmful if swallowed. Contact may cause immediate severe irritation progressing quickly to chemical burns.

Chronic Symptoms: None.

Indication of Any Immediate Medical Attention and Special Treatment Needed

If exposed or concerned, get medical advice and attention.

SECTION 5: FIREFIGHTING MEASURES

Extinguishing Media

Suitable Extinguishing Media: Water spray, fog, carbon dioxide, foam.

Unsuitable Extinguishing Media: Do not use a heavy water stream. Use of heavy stream of water may spread fire.

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Special Hazards Arising From the Substance or Mixture

Fire Hazard: Product is not flammable but during fire product can decompose and generate oxygen which can initiate or promote combustion. Strong oxidizer.

Explosion Hazard: Heated containers may build pressure, rupturing closed containers, spreading fire and increasing risk of burns and injuries.

Reactivity: Thermal decomposition generates: Corrosive vapors, acetic acid, and oxygen which supports combustion.

Advice for Firefighters

Precautionary Measures Fire: Exercise caution when fighting any chemical fire.

Firefighting Instructions: Use water spray or fog for cooling exposed containers. For major fire and large quantities, evacuate area. Fight fire from protected location or maximum distance because of risk that heated containers could rupture.

Protection During Firefighting: Do not enter fire area without proper protective equipment, including respiratory protection.

Hazardous Combustion Products: Corrosive vapors, acetic acid.

Other information: Do not allow run-off from fire fighting to enter drains or water courses. Chemical type extinguishers are not effective with peracetic acid or hydrogen peroxide.

Reference to Other Sections

Refer to section 9 for flammability properties.

SECTION 6: ACCIDENTAL RELEASE MEASURES

Personal Precautions, Protective Equipment and Emergency Procedures

General Measures: Do NOT breathe (vapors, mist, spray). Do not allow product to spread into the environment. Avoid all contact with skin, eyes, or clothing. Approach release from upwind.

For Non-Emergency Personnel

Protective Equipment: Use appropriate personal protection equipment (PPE).

Emergency Procedures: Evacuate unnecessary personnel.

For Emergency Personnel

Protective Equipment: Equip cleanup crew with proper protection.

Emergency Procedures: Ventilate area.

Environmental Precautions

Prevent entry to sewers and public waters. Avoid release to the environment.

Methods and Material for Containment and Cleaning Up

For Containment: Small spills may be flushed to an approved sewer line with generous amounts of water. For larger spills, dike well ahead of spill with non-reactive material such as sand. Cautiously neutralize spilled liquid. Spill may be neutralized with soda ash (sodium carbonate) broadcast on surface. Use 0.7 to 1lb. of soda ash for each gallon of spilled material. The resultant neutralized product will become carbon dioxide and water. Flush material with water and collect for disposal into plastic containers.

Methods for Cleaning Up: Clean up spills immediately and dispose of waste safely. Absorb spillage to prevent material damage. Do not take up in combustible material such as: saw dust or cellulosic material. Contact competent authorities after a spill.

Reference to Other Sections

See heading 8, Exposure Controls and Personal Protection.

SECTION 7: HANDLING AND STORAGE

Precautions for Safe Handling

Additional Hazards When Processed: May be corrosive to metals. Handle empty containers with care. May cause or intensify fire; oxidizer.

Handling Temperature: Not available.

Hygiene Measures: Handle in accordance with good industrial hygiene and safety procedures. Wash hands and other exposed areas with mild soap and water before eating, drinking, or smoking and again when leaving work. Do not eat, drink or smoke when using this product. Wash hands and forearms thoroughly after handling. Always wash your hands immediately after handling this product, and once again before leaving the workplace.

Conditions for Safe Storage, Including Any Incompatibilities

Technical Measures: None known.

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Storage Conditions: Store in a cool, dry, well-ventilated area. Do not store near reducing agents, fuels, organic materials, or other non-compatible materials. Do not store in direct sunlight, or near sources of ignition or heat. Product can be shipped on wooden pallets but should be stored on plastic pallets or plastic-covered pallets. Store drums in upright position only. Empty drums as thoroughly as possible. Triple rinse before disposal.

Incompatible Materials: Dirt, metals, strong bases, reducing agents, organic material, paper, wood, leather and heavy metals and their salts. May react violently with combustible materials. May react violently with finely divided metals. Never return product to original container.

Storage Temperature: Not available.

Storage Area: Store in a well-ventilated place. Keep cool. Protect from sunlight.

Special Rules on Packaging: Keep only in original container.

Specific End Use(s)

Detergent for use on food processing equipment. For professional use only.

SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

Control Parameters

Acetic acid (64-19-7)		
Mexico	OEL TWA (mg/m ³)	25 mg/m ³
Mexico	OEL TWA (ppm)	10 ppm
Mexico	OEL STEL (mg/m ³)	37 mg/m ³
Mexico	OEL STEL (ppm)	15 ppm
USA ACGIH	ACGIH TWA (ppm)	10 ppm
USA ACGIH	ACGIH STEL (ppm)	15 ppm
USA OSHA	OSHA PEL (TWA) (mg/m ³)	25 mg/m ³
USA OSHA	OSHA PEL (TWA) (ppm)	10 ppm
USA NIOSH	NIOSH REL (TWA) (mg/m ³)	25 mg/m ³
USA NIOSH	NIOSH REL (TWA) (ppm)	10 ppm
USA NIOSH	NIOSH REL (STEL) (mg/m ³)	37 mg/m ³
USA NIOSH	NIOSH REL (STEL) (ppm)	15 ppm
USA IDLH	US IDLH (ppm)	50 ppm
Ontario	OEL STEL (ppm)	15 ppm
Ontario	OEL TWA (ppm)	10 ppm
Québec	VECD (mg/m ³)	37 mg/m ³
Québec	VECD (ppm)	15 ppm
Québec	VEMP (mg/m ³)	25 mg/m ³
Québec	VEMP (ppm)	10 ppm
Hydrogen peroxide (7722-84-1)		
Mexico	OEL TWA (mg/m ³)	1.5 mg/m ³
Mexico	OEL TWA (ppm)	1 ppm
Mexico	OEL STEL (mg/m ³)	3 mg/m ³
Mexico	OEL STEL (ppm)	2 ppm
USA ACGIH	ACGIH TWA (ppm)	1 ppm
USA OSHA	OSHA PEL (TWA) (mg/m ³)	1.4 mg/m ³
USA OSHA	OSHA PEL (TWA) (ppm)	1 ppm
USA NIOSH	NIOSH REL (TWA) (mg/m ³)	1.4 mg/m ³
USA NIOSH	NIOSH REL (TWA) (ppm)	1 ppm
USA IDLH	US IDLH (ppm)	75 ppm
Ontario	OEL TWA (ppm)	1 ppm
Québec	VEMP (mg/m ³)	1.4 mg/m ³
Québec	VEMP (ppm)	1 ppm
Sulfuric acid (7664-93-9)		
Mexico	OEL TWA (mg/m ³)	1 mg/m ³

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USA ACGIH	ACGIH TWA (mg/m ³)	0.2 mg/m ³
USA OSHA	OSHA PEL (TWA) (mg/m ³)	1 mg/m ³
USA NIOSH	NIOSH REL (TWA) (mg/m ³)	1 mg/m ³
USA IDLH	US IDLH (mg/m ³)	15 mg/m ³
Ontario	OEL TWA (mg/m ³)	0.2 mg/m ³
Québec	VECD (mg/m ³)	3 mg/m ³
Québec	VEMP (mg/m ³)	1 mg/m ³
Peroxyacetic acid (79-21-0)		
USA ACGIH	ACGIH STEL (ppm)	0.4 ppm

Exposure Controls

PRODUCT as SOLD

Appropriate Engineering Controls: Ensure adequate ventilation, especially in confined areas. Ensure all national/local regulations are observed. Emergency eye wash fountains and safety showers should be available in the immediate vicinity of any potential exposure. If user operations generate fumes, vapors, gas, or spray use process enclosures, local exhaust ventilation or other engineering controls to keep worker exposure to airborne contaminants below any recommended or regulatory limits.

Personal Protective Equipment: Protective clothing. Protective goggles. Gloves. Face shield.



Materials for Protective Clothing: Chemically resistant materials and fabrics.

Hand Protection: Wear chemically resistant protective gloves.

Eye Protection: Chemical goggles or face shield.

Skin and Body Protection: Wear suitable protective clothing.

Respiratory Protection: Use NIOSH-approved air-purifying or supplied-air respirator where airborne concentrations of vapor or mist are expected to exceed exposure limits.

Thermal Hazard Protection: Wear suitable protective clothing.

Other Information: When using, do not eat, drink or smoke.

PRODUCT at USE DILUTION:

Appropriate Engineering Controls: Ensure adequate ventilation, especially in confined areas. Ensure all national/local regulations are observed. Emergency eye wash fountains and safety showers should be available in the immediate vicinity of any potential exposure. If user operations generate fumes, vapors, gas, or spray use process enclosures, local exhaust ventilation or other engineering controls to keep worker exposure to airborne contaminants below any recommended or regulatory limits.

Personal Protective Equipment: Protective goggles. Gloves.



Hand Protection: Wear protective gloves.

Eye Protection: Safety goggles.

Skin and Body Protection: None usually needed.

Respiratory Protection: None usually needed.

Thermal Hazard Protection: None usually needed.

Other Information: When using, do not eat, drink or smoke.

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

Information on Basic Physical and Chemical Properties

Physical State	: Liquid
Appearance	: Colorless
Odor	: Sharp, pungent, vinegar-like

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Odor Threshold	: Not available
pH	: <1
Relative Evaporation Rate (butylacetate=1)	: Not available
Melting Point	: -25.9°C (-14.62°F)
Freezing Point	: Not available
Boiling Point	: 99°C (210.2°F)
Flash Point	: >93.3°C (199.9°F)
Auto-ignition Temperature	: 270°C (518°F)
Decomposition Temperature	: Not available
Flammability (solid, gas)	: Not available
Lower Flammable Limit	: Not available
Upper Flammable Limit	: Not available
Vapor Pressure	: 22 mm Hg (25°C)
Relative Vapor Density at 20°C	: Not available
Specific Gravity	: 1.12
Solubility	: Complete.
Log Pow	: Not available
Log Kow	: Not available
Viscosity, Kinematic	: Not available
Viscosity, Dynamic	: Not available
Explosion Data – Sensitivity to Mechanical Impact	: Not available
Explosion Data – Sensitivity to Static Discharge	: Not available

SECTION 10: STABILITY AND REACTIVITY

Reactivity: Thermal decomposition generates: Corrosive vapors. When heated to decomposition, generates acetic acid and oxygen which supports combustion.

Chemical Stability: Shelf life is one year from date of manufacture. Oxidizer - May intensify fire.

Possibility of Hazardous Reactions: Hazardous polymerization will not occur.

Conditions to Avoid: Open flames, elevated temperatures, any source of heat, direct sunlight. Combustibles such as paper, wood, and leather. Higher temperatures will accelerate decomposition resulting in a loss of assay.

Incompatible Materials: Dirt, metals, strong bases, reducing agents, organic material, paper, wood, leather and heavy metals and their salts. May react violently with combustible materials. May react violently with finely divided metals.

Hazardous Decomposition Products: Thermal decomposition generates: Corrosive vapors, acetic acid and oxygen which supports combustion.

SECTION 11: TOXICOLOGICAL INFORMATION

Information on Toxicological Effects - Product

PRODUCT as SOLD

Acute Toxicity: Harmful if swallowed. Harmful if inhaled.

LD50 and LC50 Data: Not available.

Skin Corrosion/Irritation: Causes severe skin burns and eye damage.

Serious Eye Damage/Irritation: Causes serious eye damage.

Respiratory or Skin Sensitization: Not classified.

Germ Cell Mutagenicity: Not classified.

Teratogenicity: Not available.

Carcinogenicity: Not available.

Specific Target Organ Toxicity (Repeated Exposure): Not classified.

Reproductive Toxicity: Not classified.

Specific Target Organ Toxicity (Single Exposure): May cause respiratory irritation.

Aspiration Hazard: Not classified.

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Potential Adverse Human Health Effects and Symptoms: Harmful if inhaled. Harmful if swallowed.

Symptoms/Injuries After Inhalation: Harmful if inhaled. May cause respiratory irritation. Inhalation may cause immediate severe irritation progressing quickly to chemical burns.

Symptoms/Injuries After Skin Contact: Contact may cause immediate severe irritation progressing quickly to chemical burns.

Symptoms/Injuries After Eye Contact: Causes serious eye damage.

Symptoms/Injuries After Ingestion: Swallowing a small quantity of this material will result in serious health hazard. Harmful if swallowed. Contact may cause immediate severe irritation progressing quickly to chemical burns.

Chronic Symptoms: None.

PRODUCT at USE DILUTION:

Acute Toxicity: Harmful if swallowed.

LD50 and LC50 Data: Not available.

Skin Corrosion/Irritation: Causes mild skin irritation.

Serious Eye Damage/Irritation: Causes eye irritation.

Respiratory or Skin Sensitization: Not classified.

Germ Cell Mutagenicity: Not classified.

Teratogenicity: Not available.

Carcinogenicity: Not classified.

Specific Target Organ Toxicity (Repeated Exposure): Not classified.

Reproductive Toxicity: Not classified.

Specific Target Organ Toxicity (Single Exposure): Not classified.

Aspiration Hazard: Not classified.

Potential Adverse Human Health Effects and Symptoms: Harmful if swallowed.

Symptoms/Injuries After Inhalation: None expected under normal use.

Symptoms/Injuries After Skin Contact: Redness, irritation.

Symptoms/Injuries After Eye Contact: Redness, irritation.

Symptoms/Injuries After Ingestion: Swallowing a small quantity of this material will result in irritation.

Information on Toxicological Effects - Ingredient(s)

LD50 and LC50 Data:

Peroxyacetic acid (79-21-0)	
LD50 Oral Rat	263 mg/kg
LD50 Dermal Rabbit	1410 µl/kg
LC50 Inhalation Rat (mg/l)	0.3 mg/l (Exposure time: 1 h)
Water (7732-18-5)	
LD50 Oral Rat	>90000 mg/kg
Acetic acid (64-19-7)	
LD50 Oral Rat	3310 mg/kg
LD50 Dermal Rabbit	1060 µl/kg
LC50 Inhalation Rat (mg/l)	11.4 mg/l/4h
ATE (oral)	3310.000 mg/kg body weight
ATE (dust, mist)	11.400 mg/l/4h
Hydrogen peroxide (7722-84-1)	
LD50 Oral Rat	376 mg/kg
LD50 Dermal Rabbit	2000 mg/kg
LC50 Inhalation Rat (mg/l)	2 g/m ³ (Exposure time: 4 h)
IARC Group	3
Sulfuric acid (7664-93-9)	
LD50 Oral Rat	2140 mg/kg
LC50 Inhalation Rat (mg/l)	510 mg/m ³ (Exposure time: 2 h)
IARC Group	1

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SECTION 12: ECOLOGICAL INFORMATION

Toxicity

Ecology - General: Toxic to aquatic life.

Acetic acid (64-19-7)	
LC50 Fish 1	79 mg/l (Exposure time: 96 h - Species: Pimephales promelas [static])
EC50 Daphnia 1	65 mg/l (Exposure time: 48 h - Species: Daphnia magna [Static])
LC50 Fish 2	75 mg/l (Exposure time: 96 h - Species: Lepomis macrochirus [static])
Hydrogen peroxide (7722-84-1)	
LC50 Fish 1	16.4 mg/l (Exposure time: 96 h - Species: Pimephales promelas)
EC50 Daphnia 1	18 - 32 mg/l (Exposure time: 48 h - Species: Daphnia magna [Static])
LC50 Fish 2	18 - 56 mg/l (Exposure time: 96 h - Species: Lepomis macrochirus [static])
Sulfuric acid (7664-93-9)	
LC50 Fish 1	500 mg/l (Exposure time: 96 h - Species: Brachydanio rerio [static])

Persistence and Degradability

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Persistence and Degradability	May cause long-term adverse effects in the environment.

Bioaccumulative Potential

Millennium AL (AFCO 4388)	
Bioaccumulative Potential	Not established.

Peroxyacetic acid (79-21-0)	
BCF fish 1	(not bioaccumulative, rapid degradation)

Acetic acid (64-19-7)	
Log Pow	-0.31 (at 20°C)

Hydrogen peroxide (7722-84-1)	
BCF fish 1	(no bioaccumulation)

Sulfuric acid (7664-93-9)	
BCF fish 1	(no bioaccumulation)

Mobility in Soil Not available.

Other Adverse Effects

Other Information: Avoid release to the environment.

SECTION 13: DISPOSAL CONSIDERATIONS

Waste Disposal Recommendations: Dispose of waste material in accordance with all local, regional, national, provincial, territorial and international regulations.

Ecology – Waste Materials: This material is hazardous to the aquatic environment. Keep out of sewers and waterways.

SECTION 14: TRANSPORT INFORMATION

14.1 In Accordance with DOT

Proper Shipping Name	: HYDROGEN PEROXIDE AND PEROXYACETIC ACID MIXTURES, STABILIZED (with acids, water and not more than 6 percent peroxyacetic acid)
Hazard Class	: 5.1
Identification Number	: UN3149
Label Codes	: 5.1, 8
Packing Group	: II
ERG Number	: 140



14.2 In Accordance with IMDG

Proper Shipping Name	: HYDROGEN PEROXIDE AND PEROXYACETIC ACID MIXTURE STABILIZED (with acids, water and not more than 6 percent peroxyacetic acid)
Hazard Class	: 5.1

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Identification Number : UN3149

Packing Group : II

Label Codes : 5.1, 8

EmS-No. (Fire) : F-H

EmS-No. (Spillage) : S-Q



14.3 In Accordance with IATA

Proper Shipping Name : HYDROGEN PEROXIDE AND PEROXYACETIC ACID MIXTURE STABILIZED (with acids, water and not more than 6 percent peroxyacetic acid)

Packing Group : II

Identification Number : UN3149

Hazard Class : 5

Label Codes : 5.1, 8

ERG Code (IATA) : 5C



14.4 In Accordance with TDG

Proper Shipping Name : HYDROGEN PEROXIDE AND PEROXYACETIC ACID MIXTURE STABILIZED (with acid(s), water and not more than 6 per cent peroxyacetic acid)

Packing Group : II

Hazard Class : 5.1

Identification Number : UN3149

Label Codes : 5.1, 8



SECTION 15: REGULATORY INFORMATION

US Federal Regulations

Millennium AL (AFCO 4388)	
SARA Section 311/312 Hazard Classes	Immediate (acute) health hazard. Reactive hazard. Fire hazard. Delayed (chronic) health hazard.
Peroxyacetic acid (79-21-0)	
Listed on the United States TSCA (Toxic Substances Control Act) inventory.	
Listed on SARA Section 302 (Specific toxic chemical listings).	
Listed on SARA Section 313 (Specific toxic chemical listings).	
SARA Section 302 Threshold Planning Quantity (TPQ)	500
SARA Section 313 - Emission Reporting	1.0%
Water (7732-18-5)	
Listed on the United States TSCA (Toxic Substances Control Act) inventory.	
Acetic acid (64-19-7)	
Listed on the United States TSCA (Toxic Substances Control Act) inventory.	
Hydrogen peroxide (7722-84-1)	
Listed on the United States TSCA (Toxic Substances Control Act) inventory.	
Listed on SARA Section 302 (Specific toxic chemical listings).	
SARA Section 302 Threshold Planning Quantity (TPQ)	1000 (concentration >52%)
Sulfuric acid (7664-93-9)	
Listed on the United States TSCA (Toxic Substances Control Act) inventory.	
Listed on SARA Section 302 (Specific toxic chemical listings).	
Listed on SARA Section 313 (Specific toxic chemical listings).	
SARA Section 302 Threshold Planning Quantity (TPQ)	1000
SARA Section 313 - Emission Reporting	1.0% (acid aerosols including mists, vapors, gas, fog, and other airborne forms of any particle size)

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Benzenesulfonic acid, C10-16-alkyl derivatives, sodium salts (68081-81-2)

Listed on the United States TSCA (Toxic Substances Control Act) inventory.

US State Regulations

Peroxyacetic acid (79-21-0)

U.S. - California - Toxic Air Contaminant List (AB 1807, AB 2728)
U.S. - New Jersey - Discharge Prevention - List of Hazardous Substances
U.S. - New Jersey - Environmental Hazardous Substances List
U.S. - New Jersey - Right to Know Hazardous Substance List
U.S. - New Jersey - Special Health Hazards Substances List
U.S. - New Jersey - TCPA - Extraordinarily Hazardous Substances (EHS)
U.S. - New York - Reporting of Releases Part 597 - List of Hazardous Substances
U.S. - Pennsylvania - RTK (Right to Know) - Environmental Hazard List
U.S. - Pennsylvania - RTK (Right to Know) List

Acetic acid (64-19-7)

U.S. - New Jersey - Discharge Prevention - List of Hazardous Substances
U.S. - New Jersey - Right to Know Hazardous Substance List
U.S. - New Jersey - Special Health Hazards Substances List
U.S. - New York - Occupational Exposure Limits - TWAs
U.S. - New York - Reporting of Releases Part 597 - List of Hazardous Substances
U.S. - Pennsylvania - RTK (Right to Know) - Environmental Hazard List
U.S. - Pennsylvania - RTK (Right to Know) List
U.S. - Texas - Effects Screening Levels - Long Term
U.S. - Texas - Effects Screening Levels - Short Term

Hydrogen peroxide (7722-84-1)

U.S. - New Jersey - Discharge Prevention - List of Hazardous Substances
U.S. - New Jersey - Environmental Hazardous Substances List
U.S. - New Jersey - Right to Know Hazardous Substance List
U.S. - New Jersey - Special Health Hazards Substances List
U.S. - New York - Occupational Exposure Limits - TWAs
U.S. - New York - Reporting of Releases Part 597 - List of Hazardous Substances
U.S. - Pennsylvania - RTK (Right to Know) - Environmental Hazard List
U.S. - Pennsylvania - RTK (Right to Know) List
U.S. - Texas - Effects Screening Levels - Long Term
U.S. - Texas - Effects Screening Levels - Short Term

Sulfuric acid (7664-93-9)

U.S. - California - Proposition 65 - Carcinogens List

WARNING: This product contains chemicals known to the State of California to cause cancer.

Sulfuric acid (7664-93-9)

Strong inorganic acid mists containing sulfuric acid are present on the State of California list of Chemicals Known to the State to Cause Cancer or Reproductive Toxicity (Cal Prop 65).

U.S. - California - SCAQMD - Toxic Air Contaminants - Non-Cancer Acute
U.S. - California - SCAQMD - Toxic Air Contaminants - Non-Cancer Chronic
U.S. - California - Toxic Air Contaminant List (AB 1807, AB 2728)
U.S. - New Jersey - Discharge Prevention - List of Hazardous Substances
U.S. - New Jersey - Environmental Hazardous Substances List
U.S. - New Jersey - Right to Know Hazardous Substance List
U.S. - New Jersey - Special Health Hazards Substances List
U.S. - New York - Occupational Exposure Limits - TWAs
U.S. - New York - Reporting of Releases Part 597 - List of Hazardous Substances
U.S. - North Carolina - Control of Toxic Air Pollutants
U.S. - Pennsylvania - RTK (Right to Know) - Environmental Hazard List

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U.S. - Pennsylvania - RTK (Right to Know) List
U.S. - Texas - Effects Screening Levels - Long Term
U.S. - Texas - Effects Screening Levels - Short Term

Benzenesulfonic acid, C10-16-alkyl derivatives, sodium salts (68081-81-2)

U.S. - Texas - Effects Screening Levels - Long Term
U.S. - Texas - Effects Screening Levels - Short Term

Canadian Regulations

Peroxyacetic acid (79-21-0)

Listed on the Canadian DSL (Domestic Substances List) inventory.

Water (7732-18-5)

Listed on the Canadian DSL (Domestic Substances List) inventory.

Acetic acid (64-19-7)

Listed on the Canadian DSL (Domestic Substances List) inventory.

Hydrogen peroxide (7722-84-1)

Listed on the Canadian DSL (Domestic Substances List) inventory.

Sulfuric acid (7664-93-9)

Listed on the Canadian DSL (Domestic Substances List) inventory.

Benzenesulfonic acid, C10-16-alkyl derivatives, sodium salts (68081-81-2)

Listed on the Canadian DSL (Domestic Substances List) inventory.

This product has been classified in accordance with the hazard criteria of the Controlled Products Regulations (CPR) and the SDS contains all of the information required by CPR.

SECTION 16: OTHER INFORMATION

Revision date : 12/06/2024
Other Information : This document has been prepared in accordance with the SDS requirements of the OSHA Hazard Communication Standard 29 CFR 1910.1200.

GHS Full Text Phrases:

Acute Tox. 4 (Dermal)	Acute toxicity (dermal) Category 4
Acute Tox. 4 (Inhalation)	Acute toxicity (inhalation) Category 4
Acute Tox. 4 (Inhalation: dust, mist)	Acute toxicity (inhalation: dust, mist) Category 4
Acute Tox. 4 (Inhalation: vapour)	Acute toxicity (inhalation: vapour) Category 4
Acute Tox. 4 (Oral)	Acute toxicity (oral) Category 4
Aquatic Acute 1	Hazardous to the aquatic environment - Acute Hazard Category 1
Aquatic Acute 2	Hazardous to the aquatic environment - Acute Hazard Category 2
Aquatic Acute 3	Hazardous to the aquatic environment - Acute Hazard Category 3
Aquatic Chronic 3	Hazardous to the aquatic environment - Chronic Hazard Category 3
Carc. 1A	Carcinogenicity Category 1A
Eye Dam. 1	Serious eye damage/eye irritation Category 1
Eye Irrit. 2A	Serious eye damage/eye irritation Category 2A
Flam. Liq. 3	Flammable liquids Category 3
Met. Corr. 1	Corrosive to metals Category 1
Org. Perox. D	Organic Peroxide Category D
Ox. Liq. 1	Oxidizing liquids Category 1
Skin Corr. 1A	Skin corrosion/irritation Category 1A
Skin Irrit. 2	Skin corrosion/irritation Category 3
STOT SE 3	Specific target organ toxicity (single exposure) Category 3
H226	Flammable liquid and vapor
H242	Heating may cause a fire

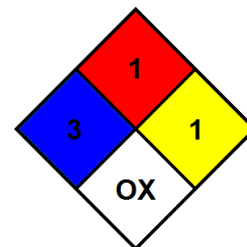
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H271	May cause fire or explosion; strong oxidizer
H290	May be corrosive to metals
H302	Harmful if swallowed
H312	Harmful in contact with skin
H314	Causes severe skin burns and eye damage
H316	Causes mild skin irritation
H318	Causes serious eye damage
H320	Causes eye irritation
H332	Harmful if inhaled
H335	May cause respiratory irritation
H350	May cause cancer
H400	Very toxic to aquatic life
H401	Toxic to aquatic life
H402	Harmful to aquatic life
H412	Harmful to aquatic life with long lasting effects

- NFPA Health Hazard** : 3 - Short exposure could cause serious temporary or residual injury even though prompt medical attention was given.
- NFPA Fire Hazard** : 1 - Must be preheated before ignition can occur.
- NFPA Reactivity** : 1 - Normally stable, but can become unstable at elevated temperatures and pressures or may react with water with some release of energy, but not violently.
- NFPA Specific Hazard** : OX - This denotes an oxidizer, a chemical which can greatly increase the rate of combustion/fire.



HMIS III Rating

- Health** : 3 Serious Hazard - Major injury likely unless prompt action is taken, and medical treatment is given.
- Flammability** : 1 - Slight Hazard.
- Physical** : 1 - Slight Hazard.

Party Responsible for the Preparation of This Document

AFCO
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This information is based on our current knowledge and is intended to describe the product for the purposes of health, safety and environmental requirements only. It should not therefore be construed as guaranteeing any specific property of the product.

North America GHS 2015 (U.S., CAN., Mex.)